Selected preparations and the Rindler time kernel

Exact source response and a semiclassical shape correction

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Abstract

The fixed-charge bilocal construction gives access to observables between controlled selected preparations. We express the single-selected functional as the second response of a normalized source state. On the charged subalgebra the source is the physical Hamiltonian. Applying the known thermal chord saddle predicts a first-order change in the curvature two-point function which cannot be absorbed into a constant rescaling of time. Exact gap-moment expansions through time degree fourteen and finite-deformation calculations test this prediction. The source identity is exact; differentiating the semiclassical saddle requires an additional uniform remainder estimate not proved here.

1 Physical question, normalization and existing input

The Susskind program asks for a quantitative comparison of derivative singlets with the bulk meson spectrum [1]. Its singlets are $\sum_i \psi_i^\dagger \partial_t^n \psi_i$. Our earlier work has identified their nonconserved components and reconstructed the actual centered bilocal in a fixed charge sector [4]. Before using its time spectrum in a bulk dictionary, we ask how the choice of preparation changes that spectrum.

Use the standard overlapping complex Gaussian p-body CAR model, the same disorder in all fields, $\hbar = 1$, string time t, and $p^2/N \to \lambda > 0$ at half filling. First take this double-scaled limit. Subsequent small-λ statements keep string time fixed. Write

$$\varepsilon = 4\lambda, \quad q = e^{-\varepsilon}, \quad C = J_s^2/\lambda, \quad e = H/\sqrt{C}, \quad K = H/(2J_s) = e/\sqrt{\varepsilon}, \quad u = 2J_s t.$$

The energy e has unit variance and K has variance $1/\varepsilon$. Let x be the normalized nonconserved curvature singlet of [3]. In its limiting tracial algebra, $\tau(x^2) = 1$ and its two-energy kernel is the q-Mehler kernel with parameter $r = q^2$. Define

$$x(u) = e^{iuK} x e^{-iuK}, \quad C_0(u) = \tau(x(u)x), \qquad C_1(u) = \sigma_1(x(u)x).$$

The previously proved single-selected restriction is

$$\sigma_1(Y) = \tau(f_q(e)Y), \qquad f_q(e) = 1 - \sqrt{q} + \sqrt{q}\, e^2, \quad Y \in W^*(e, x). \tag{1}$$

It is normalized, positive and stationary on this subalgebra. The corresponding finite-N statement is the sequential limit of actual exactly pinched observables in the specified preparation. No rate uniform in $\lambda \to 0$ is assumed.

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2 Exact source-response identities

For a centered self-adjoint A in a tracial algebra define

$$F_A(z;Y) = \frac{\tau(e^{-zA}Y)}{\tau(e^{-zA})}.$$

For bounded A this is analytic near zero. Direct differentiation, with $\tau(A) = 0$, gives

$$F_A''(0;Y) = \tau(A^2Y) - \tau(A^2)\tau(Y). \tag{2}$$

No commutation of A and Y is needed. This is the usual normalized source susceptibility, applied here to the known selected density.

For each fixed $q < 1$, K is bounded. Let

$$G_{\varepsilon,\beta}(u) = \frac{\tau(e^{-\beta K}x(u)x)}{Z_\varepsilon(\beta)}, \qquad Z_\varepsilon(\beta) = \tau(e^{-\beta K}). \tag{3}$$

Here β is dimensionless and conjugate to K; physical inverse string energy is $\beta/(2J_s)$. This is a limiting tracial Gibbs functional, not an assumed bulk Rindler state or an assertion about quenched finite-N normalization.

Proposition 1. *At every fixed $\varepsilon > 0$ and real u,*

$$C_1(u) - C_0(u) = e^{-\varepsilon/2}\varepsilon\, \partial_\beta^2 G_{\varepsilon,\beta}(u)\Big|_{\beta=0}. \tag{4}$$

Proof. Equation (1) gives the difference as $\sqrt{q}[\varepsilon\tau(K^2x(u)x) - \tau(x(u)x)]$. Apply (2) and $\tau(K^2) = 1/\varepsilon$. □

The full covariant tower contains a size field W and has the different density established in [3, 4]:

$$\rho_1 = \sqrt{q}\,(e-W)^2 + [1 - \sqrt{q}(1+\lambda)]I, \quad \tau((e-W)^2) = 1+\lambda.$$

Consequently, for the controlled integrable covariant observables,

$$\sigma_1(Y) - \tau(Y) = \sqrt{q}\,F''_{e-W}(0;Y). \tag{5}$$

At fixed q, e and W are bounded. The actual bilocal limit has all fixed moments, so the differentiation is also valid for those integrable Y. This identity retains the terms $-\tau((eW + We)Y) + \tau((W^2 - \lambda)Y)$ absent from the charged restriction. The source $e - W$ is not the physical time generator. It supplies a source response for the full bilocal, not a thermal interpretation of its generally nonstationary selected state.

3 The thermal saddle and a testable shape correction

The leading thermal matter correlator is known from large-p SYK and the semiclassical chord theory [2]. In its conventions, put $\theta = \pi v/2$; the thermal equation and real-time continuation are

$$\beta = \frac{2\theta}{\cos\theta}, \qquad g_{\beta,d}(u) = \left[\frac{\cos\theta}{\cosh(u\cos\theta + i\theta)}\right]^{2d}. \tag{6}$$

The Hamiltonian in these units is K, whose variance is $1/\varepsilon$. Our curvature channel has $d = 2$ because its Hamiltonian crossing weight is q^2, not because its bulk mass has been identified. Equation (6) is the unsplit thermal correlator; a midpoint thermal correlator has different source derivatives.

Since $\theta = \beta/2 + O(\beta^3)$, with $T = \tanh u$,

$$\log g_{\beta,d} = -2d\log\cosh u - id\beta T + \frac{d\beta^2}{4}(uT - T^2) + O(\beta^3).$$

Therefore

$$\partial_\beta^2 g_{\beta,d}\Big|_0 = \frac{d}{2}\operatorname{sech}^{2d} u\,[u\tanh u - (1+2d)\tanh^2 u]. \tag{7}$$

Combining (7) with the exact (4) predicts

$$\frac{C_1(u) - C_0(u)}{\varepsilon} \longrightarrow R(u) := \operatorname{sech}^4 u\,[u\tanh u - 5\tanh^2 u]. \tag{8}$$

The precise sufficient hypothesis is convergence of $\partial_\beta^2 G_{\varepsilon,\beta}|_0$ to $\partial_\beta^2 g_{\beta,2}|_0$, locally uniformly in u. A locally uniform analytic saddle expansion in a complex β neighborhood would imply it by Cauchy's formula. Pointwise thermal convergence alone does not. We do not prove this differentiated remainder bound in this note, and label (8) as a semiclassical prediction rather than a new all-time convergence theorem.

Nevertheless the short-time coefficient has an independent exact check. Mehler conditional moments give

$$-(C_1 - C_0)''(0) = \frac{e^{-\varepsilon/2}(1+e^{-\varepsilon})(1-e^{-2\varepsilon})^2}{\varepsilon} = 8\varepsilon + O(\varepsilon^2). \tag{9}$$

The prediction has $R(u) = -4u^2 + 11u^4 + O(u^6)$. It cannot be a constant time recalibration: $au(\operatorname{sech}^4 u)' = -4au^2 + (28a/3)u^4 + O(u^6)$. Matching the quadratic term fixes $a = 1$, which fails at the quartic term. An overall amplitude is fixed by $C_1(0) = C_0(0) = 1$.

4 Independent coefficients and finite-deformation evidence

Let Q_j be the monic unit-variance q-Hermite polynomials, $e^a = \sum_j b_{aj} Q_j(e)$. The exact two-energy moments are

$$T_{ab}(q) = \sum_j b_{aj} b_{bj} [j]_q!\, q^{2j}.$$

Thus the selected-minus-trace gap moment before physical time rescaling is

$$D_n(q) = \sum_{a=0}^{n} (-1)^{n-a} \binom{n}{a} [T_{a+2,n-a}(q) - T_{a,n-a}(q)].$$

All quantities here are integer polynomials in q. If $D_{2k}(q) = \sum_j d_j q^j$, the coefficient of ε^ℓ in $e^{-\varepsilon/2} D_{2k}(e^{-\varepsilon})$ is exactly

$$\frac{1}{\ell!} \sum_j d_j (-j - 1/2)^\ell.$$

The physical time coefficient includes $(-1)^k/[(2k)!\varepsilon^k]$. For $2k = 0, 2, \ldots, 14$, all 36 terms with $\ell \le k$ vanish exactly. The first surviving coefficients reproduce

$$R(u) = -4u^2 + 11u^4 - \frac{769}{45}u^6 + \frac{6236}{315}u^8 - \frac{1201}{63}u^{10} + \frac{7551623}{467775}u^{12} - \frac{529250077}{42567525}u^{14} + O(u^{16}).$$

These are finite exact checks, not a proof for all Taylor orders.

For real time use the Jacobi matrix T with off-diagonal $\sqrt{[n]_q}$ and $D = \mathrm{diag}(q^{2n})$. With $a = e^{iuT/\sqrt{\varepsilon}}|0\rangle$ and $b = e^{iuT/\sqrt{\varepsilon}}|2\rangle$, the exact infinite-matrix formulas inherited from [3] are

$$C_0 = \langle a, Da\rangle, \qquad C_1 - C_0 = \sqrt{q(1+q)}\langle b, Da\rangle.$$

The difference is computed directly. On the seven-point grid $u = 0, 0.25, \ldots, 1.5$ the finite-deformation results are

ε	Jacobi dimensions	$\max\lvert(C_1 - C_0)/\varepsilon - R\rvert$
0.08	214,278	0.077469
0.04	364,428	0.040022
0.02	664,728	0.020335
0.01	1264,1328	0.010249

The first dimension uses $\lceil 12/\varepsilon \rceil + 64$. The displayed differences measure finite-deformation bias, not numerical error. The parameter λ equals $\varepsilon/4$.

5 Error accounting and the next construction

For $z = 2|u|/\sqrt{\varepsilon(1-q)}$, a common exact-arithmetic packet truncation bound inherited from [3] is

$$4[1+\sqrt{q(1+q)}]e^{z}\frac{z^{L-2}}{(L-2)!}.$$

The saved parameters satisfy $z^2 \le L(L-1)$; all nonzero-time bounds are below 10^{-25}. This bounds truncation, not floating roundoff. We compare L and L+64, retain their difference after division by ε, check packet normalization, and at $\varepsilon = 0.08$ compare sparse exponential action against an independent symmetric tridiagonal eigensolver. All numerical comparisons use float64 and empirical tolerances, not intervals.

The largest summand-cancellation ratio is about 414. For example, at $\varepsilon = 0.01, u = 0.25$ the coefficient differs from the prediction by about 2.68%, despite negligible cutoff change. At $u = 0.5$ its error is about 1.98%. These differences must not be relabeled as numerical uncertainty or omitted by an extrapolation.

The physical outcome is a source-response method for handling preparation dependence while retaining the actual microscopic observable. Under the differentiated saddle hypothesis, the trace and selected curvature have the same leading Rindler kernel, but a distinct first correction. This is state response, not a change in an invariant particle mass.

The next analytic step is to control the twice-differentiated thermal q-Hermite integral near $\beta = 0$, with fixed compact u. The full bilocal additionally needs the mixed e,W source in (5); using (4) there would drop real neutral terms. Spatial translations, mass generators, arbitrary selected interlacings and finite-G dynamics remain separate unfinished parts of the intended dictionary. This stage introduces no new Git publication or issue reply.

References

[1] S. Miyashita, Y. Sekino and L. Susskind, *Holograms and Standard Models*, arXiv:2607.05678v2, Secs. 4.1–4.3. The mass-spectrum comparison is an open quantitative test; a boost kernel alone does not determine that spectrum.

[2] H. W. Lin and D. Stanford, *A symmetry algebra in double-scaled SYK*, arXiv:2307.15725, Eqs. (137)–(138), Sec. 4.3 and Appendix H, Eqs. (409)–(410). Their chord deformation parameter is our ε.

[3] A. Day and Chengheng, *Single-selected covariant density and curvature dynamics*, 8 November 2026; `manuscript/susskind-slater-selected-state/main.tex`.

[4] A. Day and Chengheng, *Fixed-charge bilocals and selected preparations*, 8 November 2026; `manuscript/susskind-sector-bilocal/main.tex`.

[5] A. Day and Chengheng, bulk meson target and translation notes, 6 November 2026; `notes/susskind-bulk-target.md`, `notes/susskind-translations.md`.